

Developing Open Standards for the Metaverse

For 22 years now, the Khronos Group has been silently working on open standards for a range of technologies that help build the metaverse, including 3D graphics, parallel computation, vision acceleration, VR, AR and ML. In June this year, it launched the Metaverse Standards Forum, together with big names like Meta, Microsoft, NVIDIA, Qualcomm, OpenAR Cloud, the Open Geospatial Consortium, the World Wide Web Consortium, and so on. The forum attracted close to 1000 members within a month.

The forum, hosted by Khronos, is open to any company, standards organisation, or university at no charge. The activities of the forum will be directed by the needs and interests of its members, and will hopefully cover the entire spectrum of technologies needed by the metaverse, including 3D assets and rendering, AR and VR, user created content, avatars, identity management, privacy, legal issues and financial transactions. To know more, or to join the forum, head to <https://metaverse-standards.org/>.

to transfer drug money or stolen cryptos!

In a recent tweet, Alex Bornyakov, Deputy Minister of Digital Transformation of Ukraine, noted that crypto is essential for Ukraine's defence. "Despite the 'bear market,' crypto has become an essential tool of Ukraine's defence, providing flexibility and speed that literally saved our soldiers' lives. Every crypto winter leads to a crypto spring, but the industry is here to stay," he wrote.

According to various media reports, since the beginning of the Russia-Ukraine war, Ukraine has raised over \$135 million through crypto donations. Various artists and organisations have also helped the country raise funds through the sale of non-fungible tokens (NFTs). Ukraine's Meta History Museum of War platform, for example, raised \$1.3 million from NFT sales. The government hopes to use the fund to restore the destroyed physical monuments in the country.

India, on the other hand, is not much in favour of cryptocurrencies. Earlier this year, the Finance Minister announced the imminent launch of the digital rupee—a digital equivalent of the Indian rupee. But this is not to be confused with unregulated cryptocurrencies.

While fielding questions on the regulation of crypto, in the *Lok Sabha* in July, Finance Min-

ister Nirmala Sitharaman said the government would need international consensus and collaboration to implement any legislation on cryptocurrencies. "In view of the concerns expressed on the destabilising effect of cryptocurrencies on a country's monetary and fiscal stability, the RBI has recommended framing legislation on this sector. The RBI believes that cryptocurrencies should be prohibited," she said. Since cryptocurrencies are technically borderless, efforts to regulate or ban them will be effective only if there is a global scale evaluation, collaboration, and consensus.

In a recent media interview, Subhash Chandra Garg, former Finance and Economic Affairs Secretary, expressed his opinion that the cloud around crypto can be removed through proper dialogues between the industry and the government. "So far only the Reserve Bank of India (RBI) has shown a sort of distress about cryptocurrency. There is no opposition in the system against blockchain or other technologies. RBI is a reasonably powerful institution in the country. There is a lurking suspicion about 'crypto-thing.' This made the government take a more sledge-hammer approach. If the government goes and tries to kill crypto, we all will be losers. But this discomfort can only be removed when a dialogue takes place between the industry and the Centre," he said.

The government's pessimistic view of cryptocurrencies is also understandable. There is no way to judge the true value of cryptocurrencies, and there is a real concern about it being used by unscrupulous people to transact even huge sums of money without the government's knowledge. This can be used to fund drugs and terrorism, and can eventually even destabilise an economy. Looks like we have to wait for a global consensus on this. Till then, take it all with a pinch of salt.

Here is a quick round-up of some interesting metaverse-related developments...

Automakers in the metaverse

Some automakers like Mahindra & Mahindra were early birds in the metaverse. July saw a few more carmakers, like Suzuki and Volvo, planting their foot firmly in this new world.

Maruti Suzuki India launched the NEXAverse, a metaverse initiative to take its premium cars to the tech-savvy generation, and unveiled the Grand Vitara in it. This enables users to experience the made-in-India, mid-size sport utility vehicle (SUV) in a 3D virtual space. Maruti claims it is the first carmaker in the world to let users book their car through the metaverse platform itself.

Soon after, Volvo Cars launched the XC40 Recharge, a pure electric, locally-assembled luxury car, on VolvoVerse, its new metaverse platform. It was an exclusive metaverse launch, conceptualised by Mindshare India. The car will be sold only online and is available for booking on the company's website. This move is believed to be in line with the company's sustainability goals.

Metaverse one of the fastest growing verticals for Tech Mahindra

Tech Mahindra announced its metaverse practice in February this